

Algorithm Name

Online Gradient Descent for Strongly Convex Online Functions (OGD-SC)

Mathematical Formulation

We consider an online sequence of loss functions $\{f_t : \mathcal{W} \rightarrow \mathbb{R}\}_{t=1}^T$ revealed in rounds $t = 1, 2, \dots, T$. - Decision set: $\mathcal{W} \subseteq \mathbb{R}^d$ is a nonempty closed convex set. - At round t , the learner picks $w_t \in \mathcal{W}$, then observes f_t and incurs loss $f_t(w_t)$. A subgradient (or gradient) $g_t \in \partial f_t(w_t)$ is revealed and used to update. - Update rule (projected OGD):

$$w_{t+1} = \Pi_{\mathcal{W}}(w_t - \eta_t g_t),$$

where $\eta_t > 0$ is a step size and $\Pi_{\mathcal{W}}$ denotes Euclidean projection onto $\mathcal{W}$:

$$\Pi_{\mathcal{W}}(x) \triangleq \arg \min_{w \in \mathcal{W}} \|w - x\|_2.$$

- Step-size schedule for strongly convex losses:

$$\eta_t = \frac{1}{\lambda t}, \quad t \geq 1,$$

where $\lambda > 0$ is the strong convexity parameter.

Performance is measured by the regret after T rounds:

$$R_T \triangleq \sum_{t=1}^T f_t(w_t) - \min_{w \in \mathcal{W}} \sum_{t=1}^T f_t(w).$$

Pseudocode

Algorithm 1 OGD-SC: Online Gradient Descent for Strongly Convex Losses

- 1: **Input:** Strong convexity parameter $\lambda > 0$, horizon T , decision set $\mathcal{W} \subseteq \mathbb{R}^d$
 - 2: **Initialize:** Pick any $w_1 \in \mathcal{W}$
 - 3: **for** $t = 1$ to T **do**
 - 4: Receive (or evaluate) loss $f_t : \mathcal{W} \rightarrow \mathbb{R}$
 - 5: Compute a subgradient $g_t \in \partial f_t(w_t)$
 - 6: Set step size $\eta_t = 1/(\lambda t)$
 - 7: Update $w_{t+1} = \Pi_{\mathcal{W}}(w_t - \eta_t g_t)$
 - 8: **end for**
 - 9: **Output:** Sequence $\{w_t\}_{t=1}^T$ (or an average if desired)
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Dry Run Example

We apply the basic (unconstrained) gradient update to the single function $f(w) = (w - 3)^2$ in one dimension with $w_1 = 0$, constant step size $\eta = 0.1$, for 3 iterations. The gradient is $g(w) = \nabla f(w) = 2(w - 3)$. - Iteration $t = 1$: - Current $w_1 = 0$ - Gradient: $g_1 = 2(0 - 3) = -6$ - Update: $w_2 = w_1 - \eta g_1 = 0 - 0.1(-6) = 0.6$ - Objective: $f(w_1) = (0 - 3)^2 = 9$ - Iteration $t = 2$: - Current $w_2 = 0.6$ - Gradient: $g_2 = 2(0.6 - 3) = -4.8$ - Update: $w_3 = 0.6 - 0.1(-4.8) = 1.08$ - Objective: $f(w_2) = (0.6 - 3)^2 = 5.76$ - Iteration $t = 3$: - Current $w_3 = 1.08$ - Gradient: $g_3 = 2(1.08 - 3) = -3.84$ - Update: $w_4 = 1.08 - 0.1(-3.84) = 1.464$ - Objective: $f(w_3) = (1.08 - 3)^2 = 3.6864$

Assumptions

We assume the following throughout the analysis: - A1 (Convex domain): $\mathcal{W} \subseteq \mathbb{R}^d$ is a nonempty closed convex set. - A2 (Strong convexity): Each $f_t : \mathcal{W} \rightarrow \mathbb{R}$ is λ -strongly convex with respect to the Euclidean norm, i.e., for all $u, v \in \mathcal{W}$ and any $g_t(v) \in \partial f_t(v)$,

$$f_t(u) \geq f_t(v) + \langle g_t(v), u - v \rangle + \frac{\lambda}{2} \|u - v\|^2.$$

- A3 (Bounded subgradients): There exists $G > 0$ such that for all t and all $w \in \mathcal{W}$, any subgradient $g \in \partial f_t(w)$ satisfies $\|g\| \leq G$. - A4 (Step size): $\eta_t = \frac{1}{\lambda t}$.

Main Convergence Theorem

Theorem 1 (Logarithmic Regret for Strongly Convex Online Functions). *Under Assumptions A1–A4, the regret of OGD-SC satisfies*

$$R_T \leq \frac{G^2}{2\lambda} (1 + \ln T), \quad \text{for all } T \geq 1.$$

Theoretical Properties

- Stability: The update satisfies $\|w_{t+1} - w_t\| \leq \eta_t \|g_t\| \leq \eta_t G = \frac{G}{\lambda t}$, which vanishes as t grows, implying stability of the iterates. - Hyperparameter sensitivity: The schedule $\eta_t = \frac{1}{\alpha t}$ yields $R_T \leq \frac{G^2}{2\alpha} (1 + \ln T)$ provided $\alpha \leq \lambda$ (see Special Cases), with the best constant at $\alpha = \lambda$. - Comparison to convex (not strongly convex) case: For merely convex Lipschitz losses, OGD achieves $R_T = O(\sqrt{T})$ with an appropriate constant step size or $\eta_t \propto 1/\sqrt{t}$; strong convexity improves this to $O(\log T)$.

Discussion

The algorithm does not target convergence to a fixed point; instead, it minimizes regret against the best fixed comparator in hindsight. Strong convexity sharpens the per-round stability and enables a telescoping argument that removes dependence on the domain diameter, yielding a logarithmic regret bound. The bound holds against an arbitrary (possibly adversarial) sequence of strongly convex losses satisfying the gradient bound.

Preliminaries (Definitions and Lemmas)

Definition 1 (Regret). *For a sequence $\{f_t\}_{t=1}^T$ and actions $\{w_t\}_{t=1}^T$, the regret is*

$$R_T = \sum_{t=1}^T f_t(w_t) - \min_{w \in \mathcal{W}} \sum_{t=1}^T f_t(w).$$

Let $w^ \in \arg \min_{w \in \mathcal{W}} \sum_{t=1}^T f_t(w)$ be any minimizer.*

Lemma 1 (Euclidean Projection Inequality). *Let $\Pi_{\mathcal{W}}$ be the Euclidean projection onto a nonempty closed convex set $\mathcal{W}$. For any $x \in \mathbb{R}^d$ and any $y \in \mathcal{W}$,*

$$\|\Pi_{\mathcal{W}}(x) - y\|^2 \leq \|x - y\|^2 - \|x - \Pi_{\mathcal{W}}(x)\|^2 \leq \|x - y\|^2.$$

Proof. Let $p = \Pi_{\mathcal{W}}(x)$. By the first-order optimality condition of the projection onto a convex set, we have $\langle x - p, y - p \rangle \leq 0$ for all $y \in \mathcal{W}$.

$$\|x - y\|^2 = \|x - p + p - y\|^2 = \|x - p\|^2 + \|p - y\|^2 + 2\langle x - p, p - y \rangle. \quad [\text{Step 1: expand square}]$$

Using $\langle x - p, y - p \rangle \leq 0$ gives $\langle x - p, p - y \rangle \geq 0$. Hence

$$\|x - y\|^2 \geq \|x - p\|^2 + \|p - y\|^2. \quad [\text{Step 2: drop nonnegative cross term}]$$

$$\text{Rearranging yields } \|p - y\|^2 \leq \|x - y\|^2 - \|x - p\|^2 \leq \|x - y\|^2. \quad [\text{Step 3: algebraic rearrangement}]$$

□

Lemma 2 (One-Step Progress Inequality for OGD). *Let $w_{t+1} = \Pi_{\mathcal{W}}(w_t - \eta_t g_t)$, with any $g_t \in \mathbb{R}^d$ and $\eta_t > 0$. Then for any $w^* \in \mathcal{W}$,*

$$\langle g_t, w_t - w^* \rangle \leq \frac{\|w_t - w^*\|^2 - \|w_{t+1} - w^*\|^2}{2\eta_t} + \frac{\eta_t}{2} \|g_t\|^2.$$

Proof. Let $x_t = w_t - \eta_t g_t$. Then by Lemma 1 with $x = x_t$, $y = w^*$, and $p = w_{t+1} = \Pi_{\mathcal{W}}(x_t)$,

$$\|w_{t+1} - w^*\|^2 \leq \|x_t - w^*\|^2. \quad [\text{Step 4: apply projection inequality}]$$

Expand $\|x_t - w^*\|^2$:

$$\|x_t - w^*\|^2 = \|w_t - \eta_t g_t - w^*\|^2 = \|w_t - w^*\|^2 - 2\eta_t \langle g_t, w_t - w^* \rangle + \eta_t^2 \|g_t\|^2. \quad [\text{Step 5: expand square}]$$

Combine the two displays:

$$\|w_{t+1} - w^*\|^2 \leq \|w_t - w^*\|^2 - 2\eta_t \langle g_t, w_t - w^* \rangle + \eta_t^2 \|g_t\|^2. \quad [\text{Step 6: transitivity}]$$

Rearrange to isolate $\langle g_t, w_t - w^* \rangle$:

$$2\eta_t \langle g_t, w_t - w^* \rangle \leq \|w_t - w^*\|^2 - \|w_{t+1} - w^*\|^2 + \eta_t^2 \|g_t\|^2. \quad [\text{Step 7: algebra}]$$

Divide by $2\eta_t > 0$ to conclude:

$$\langle g_t, w_t - w^* \rangle \leq \frac{\|w_t - w^*\|^2 - \|w_{t+1} - w^*\|^2}{2\eta_t} + \frac{\eta_t}{2} \|g_t\|^2. \quad [\text{Step 8: divide by } 2\eta_t]$$

□

Lemma 3 (Strong Convexity Upper Bound). *If f_t is λ -strongly convex and $g_t \in \partial f_t(w_t)$, then for any $w^* \in \mathcal{W}$,*

$$f_t(w_t) - f_t(w^*) \leq \langle g_t, w_t - w^* \rangle - \frac{\lambda}{2} \|w_t - w^*\|^2.$$

Proof. By strong convexity with $u = w^*$ and $v = w_t$,

$$f_t(w^*) \geq f_t(w_t) + \langle g_t, w^* - w_t \rangle + \frac{\lambda}{2} \|w^* - w_t\|^2. \quad [\text{Step 9: apply definition}]$$

Rearrange:

$$f_t(w_t) - f_t(w^*) \leq \langle g_t, w_t - w^* \rangle - \frac{\lambda}{2} \|w_t - w^*\|^2. \quad [\text{Step 10: move terms, use symmetry of norm}]$$

□

Lemma 4 (Harmonic Series Bound). *For all integers $T \geq 1$, $\sum_{t=1}^T \frac{1}{t} \leq 1 + \ln T$.*

Proof. For $t \geq 2$, $\int_{t-1}^t \frac{1}{x} dx = \ln t - \ln(t-1) \leq \frac{1}{t-1}$. Hence

$$\sum_{t=2}^T \frac{1}{t} \leq \int_1^T \frac{1}{x} dx = \ln T. \quad [\text{Step 11: integral comparison}]$$

Add $\frac{1}{1} = 1$ to both sides to conclude the claim. [Step 12: add $t = 1$ term]

□

Main Proof (Step-by-Step Derivation)

Proof of Theorem 1. Fix $w^* \in \arg \min_{w \in \mathcal{W}} \sum_{t=1}^T f_t(w)$. For each t , apply Lemma 3 and then Lemma 2:

$$\begin{aligned} f_t(w_t) - f_t(w^*) &\leq \langle g_t, w_t - w^* \rangle - \frac{\lambda}{2} \|w_t - w^*\|^2 \quad [\text{Step 13: Lemma 3}] \\ &\leq \frac{\|w_t - w^*\|^2 - \|w_{t+1} - w^*\|^2}{2\eta_t} + \frac{\eta_t}{2} \|g_t\|^2 - \frac{\lambda}{2} \|w_t - w^*\|^2. \quad [\text{Step 14: Lemma 2}] \end{aligned}$$

Sum the inequality over $t = 1$ to T :

$$\sum_{t=1}^T (f_t(w_t) - f_t(w^*)) \leq \sum_{t=1}^T \left[\frac{\|w_t - w^*\|^2 - \|w_{t+1} - w^*\|^2}{2\eta_t} - \frac{\lambda}{2} \|w_t - w^*\|^2 \right] + \frac{1}{2} \sum_{t=1}^T \eta_t \|g_t\|^2. \quad [\text{Step 15: s}]$$

Introduce $u_t \triangleq \|w_t - w^*\|^2$ and $c_t \triangleq \frac{1}{\eta_t}$. Then the bracketed sum equals

$$\begin{aligned} \frac{1}{2} \sum_{t=1}^T (c_t(u_t - u_{t+1}) - \lambda u_t) &= \frac{1}{2} \left[c_1 u_1 - c_T u_{T+1} + \sum_{t=2}^T (c_t - c_{t-1}) u_t - \lambda \sum_{t=1}^T u_t \right] \quad [\text{Step 16: telescoping id}] \\ &= \frac{1}{2} \left[(c_1 - \lambda) u_1 - c_T u_{T+1} + \sum_{t=2}^T ((c_t - c_{t-1}) - \lambda) u_t \right]. \quad [\text{Step 17: regroup ter}] \end{aligned}$$

Now specialize to $\eta_t = \frac{1}{\lambda t}$ so that $c_t = \lambda t$ is affine in t :

$$c_1 - \lambda = \lambda - \lambda = 0, \quad c_t - c_{t-1} - \lambda = \lambda - \lambda = 0 \text{ for all } t \geq 2. \quad [\text{Step 18: compute differences}]$$

Therefore the entire bracketed sum reduces to

$$\frac{1}{2} [-c_T u_{T+1}] \leq 0, \quad \text{since } c_T u_{T+1} \geq 0. \quad [\text{Step 19: nonnegativity implies upper bound by 0}]$$

Discarding this nonpositive term yields

$$\sum_{t=1}^T (f_t(w_t) - f_t(w^*)) \leq \frac{1}{2} \sum_{t=1}^T \eta_t \|g_t\|^2. \quad [\text{Step 20: drop nonpositive contribution}]$$

Using Assumption A3, $\|g_t\| \leq G$ for all t , and $\eta_t = \frac{1}{\lambda t}$,

$$\sum_{t=1}^T (f_t(w_t) - f_t(w^*)) \leq \frac{1}{2} \sum_{t=1}^T \frac{G^2}{\lambda t} = \frac{G^2}{2\lambda} \sum_{t=1}^T \frac{1}{t}. \quad [\text{Step 21: bound gradients and plug step sizes}]$$

Apply Lemma 4 to bound the harmonic series:

$$\sum_{t=1}^T (f_t(w_t) - f_t(w^*)) \leq \frac{G^2}{2\lambda} (1 + \ln T). \quad [\text{Step 22: Lemma 4}]$$

Since the left-hand side equals the regret R_T by definition, the theorem follows. [Step 23: identify regret]

□

Rate Derivation (With Constants)

With $\eta_t = \frac{1}{\lambda t}$ and $\|g_t\| \leq G$,

$$R_T \leq \frac{1}{2} \sum_{t=1}^T \frac{G^2}{\lambda t} \leq \frac{G^2}{2\lambda} (1 + \ln T),$$

where: - G is the uniform bound on subgradient norms (A3), - λ is the strong convexity parameter (A2), - The constant 1 in $1 + \ln T$ comes from the $t = 1$ harmonic term.

Special Cases

- Unconstrained case ($\mathcal{W} = \mathbb{R}^d$): The projection becomes identity, i.e., $w_{t+1} = w_t - \eta_t g_t$. Lemma 2 holds with equality at Step 4, and the same regret bound follows. - Misspecified step size $\eta_t = \frac{1}{\alpha t}$ with $\alpha \leq \lambda$: Then $c_t = \alpha t$ and $c_t - c_{t-1} - \lambda = \alpha - \lambda \leq 0$. In Steps 16–20, the bracketed sum is ≤ 0 and can be dropped, yielding

$$R_T \leq \frac{G^2}{2\alpha} \sum_{t=1}^T \frac{1}{t} \leq \frac{G^2}{2\alpha} (1 + \ln T),$$

which is largest when α is much smaller than λ ; the best constant occurs at $\alpha = \lambda$. - Overaggressive schedule $\eta_t = \frac{1}{\alpha t}$ with $\alpha > \lambda$: Then $c_t - c_{t-1} - \lambda = \alpha - \lambda > 0$, and the bracketed sum in Step 17 contributes a positive term that requires an a priori bound on $\|w_t - w^*\|^2$ (e.g., via bounded $\mathcal{W}$). Without such a bound, the logarithmic guarantee may fail; thus choosing $\alpha \leq \lambda$ is recommended for diameter-free regret.